# Pranav Joshi

pranav.joshi@alumni.iitgn.ac.in | pranavajoshi04@gmail.com | +91 8698955551
GitHub | LinkedIn | Academic Portfolio | Medium

Engineer and researcher specializing in FPGA hardware acceleration, ultrasonic image reconstruction, real-time signal processing, and statistical modeling. Proven expertise in deriving and implementing mathematical and hardware-software co-design frameworks; from high-throughput beamforming architectures to physiological time-series modeling.

## EDUCATION

**Bachelor of Technology in Computer Science and Engineering** **Graduated: July 2026**
*Indian Institute of Technology Gandhinagar (IITGN)* *CGPA: 9.59 / 10*
**Relevant Coursework:** Signals & Systems, Digital Systems, Computer Architecture, Computer Vision, Machine Learning, Data Science, Biostatistics, Computational Physics, Databases.

## PUBLICATIONS, POSTERS & PREPRINTS

- **Pranav Joshi**, Abhilipsa Sahoo, Joycee Mekie and Himanshu Shekhar, "Novel Transducer Domain Partitioning Architecture for FPGA-Based Total Focusing Method Ultrasonic Imaging", *Accepted Poster at IEEE Symposium on Acoustic and Ultrasonic Systems (IEEE SAUS)*, 2026. *(Full manuscript in preparation).*
- **Pranav Joshi, "Sampling-Based Estimation of Jaccard Containment and Similarity"**, *arXiv preprint arXiv:2507.10019*, 2025.
- **Pranav Joshi** and Ashutosh Srivastava, "Characterizing Social Jet-Lag Dynamics from Consumer Wearables", *Manuscript in preparation, Computational Biology Lab (CLOCK), IITGN*, 2026.

## RESEARCH & PROFESSIONAL EXPERIENCE

**Independent Research: Co-Evolutionary Optimization for Sparse-Array TFM** *Ongoing*
*Hardware-Friendly Ultrasonic Beamforming & Design Space Exploration*

- Developing a time-complexity optimized Cooperative Coevolutionary Genetic Algorithm (CCGA) to jointly evolve sparse transducer array layouts and non-linear partial-image activation functions.
- Replacing resource-heavy $p^{\text{th}}$-root and $p^{\text{th}}$-power coherence operations with learned symbolic and linear regression models over FPGA-implementable basis functions to amplify constructive interference and suppress noise with minimal LUT/DSP overhead.

**FPGA Research Contractor** *May – July 2026*
*MUSE Lab, IIT Gandhinagar* Gandhinagar, India

- Ramped up on ultrasound beamforming theory and background material; documented learnings in a comprehensive Medical Imaging Systems Blog
- Synthesized and implemented an end-to-end Total Focusing Method (TFM) hardware accelerator for Non-Destructive Testing (NDT) ultrasonic imaging on Nexys 4 DDR and Xilinx AC701 FPGA boards.
- Achieved real-time reconstruction at **30 fps for a** $128 \times 128$ **pixel grid** on a single FPGA via novel transducer domain partitioning, dynamic programming, deep pipelining, and AXI-stream batching.
- Developed a simulation and reconstruction library featuring Software-Assisted Configuration, storing pre-computed delay laws in DDR3 memory to bypass on-chip CORDIC latency and support complex acoustic geometries (e.g., angled wedges).
- Engineered a custom link-layer protocol over RMII/RGMII Ethernet to minimize OS context-switching overhead.
- Completed technology transfer to industry partner Modsonic Instruments Manufacturing Co. Pvt. Ltd.

**Research Intern** *May – June 2026*
*Computational Biology Lab (CLOCK), IIT Gandhinagar* Gandhinagar, India

- **Physiological Signal Conditioning:** Proposed a statistical framework to quantify intra-individual Social Jet-Lag (SJL) from high-granularity wearable sensor streams (Ultrahuman). Engineered an unsupervised pipeline using 2D histograms, $k$-means clustering, and unpaired $t$-tests to isolate and filter firmware-induced motion artifacts in Heart Rate (HR) and Heart Rate Variability (HRV).
- **Metric Formulation:** Formulated the Heart-Rate-Autocorrelation (HRA) metric and burstiness index to characterize acute SJL events, controlling for confounding variables via Ordinary Least Squares (OLS) regression and validating superiority over traditional Sleep Midpoint tracking.
- **Time-Series Forecasting:** Implemented a Cosinor autoregressive harmonic regression model to extract circadian phase/amplitude parameters, and benchmarked it against a custom multi-modal Transformer architecture for forecasting physiological trajectories.

**Data Engineer (July 2026 – Present) & Data Intern (May – July 2025)** *May 2025 – Present*
*Texas Instruments* Bengaluru, India

- **AI-Assisted Hardware System Design (Data Engineer):** Developing AI-driven Block Diagram generation tools for `ti.com` to automate component selection and accelerate high-level end-equipment architecture design.
- **Large-Scale Statistical Retrieval (Data Intern):** Engineered a metadata-free Retrieval-Augmented Generation (RAG) and multi-agent Text-to-SQL pipeline scaling across thousands of columns and $\sim 10^9$ data points in enterprise Oracle databases. Overcoming rate-limited schema matching directly motivated the mathematical formulation of the arXiv Jaccard Containment estimator.
- **Recognition:** Awarded a Pre-Placement Offer (PPO) for outstanding performance during the internship, leading to the current Data Engineer role.

## SELECTED ACADEMIC PROJECTS

**Numerical Modeling & Simulation of Wave and Dynamical Systems** | *PH509, IITGN* *2026*

- Developed $\mathcal{O}(N)$ and $\mathcal{O}(MN)$ Time-Dependent Schrödinger Equation (TDSE) wave-propagation solvers using Split-Operator and Alternating Direction Implicit (ADI) tridiagonal matrix methods.
- Performed von Neumann stability analysis on Space-Discretized Leap-Frog (SDLF) schemes to prevent numerical divergence.
- Built adaptive Position-Verlet Leapfrog integrators for 2-body orbital mechanics, verifying conservation laws.

**Person Velocity Estimation via Projective Geometry & Computer Vision** | *IITGN* *2024*

- Integrated YOLO-based object and head tracking pipelines with native projective geometry homography transformations in OpenCV to estimate real-time physical velocity from monocular video feeds.

**Automatic Differentiation Engine & Classical ML from Scratch** | *ES335, IITGN* *2023*

- Engineered an automatic differentiation (autograd) engine supporting dynamic computational graphs, backpropagation, and modular neural network layers.
- Built custom decision tree and random forest ensembles driven by Shannon Entropy and RMSE minimization heuristics.

**Open Source Contributions to Computational Group Theory** | *IITGN* *Jan – Jun 2024*

- Collaborated with the author of two research papers on group-theoretic algorithms to implement algebraic routines in the GAP and SageMath symbolic computation ecosystems.

**Quantized Auto-Encoder Neural Network on FPGA** | *Digital Systems, IITGN* *2023*

- Designed and synthesized a feed-forward quantized auto-encoder neural network at the Register-Transfer Level (RTL) using SystemVerilog on an FPGA development board.

**AlgoLang: Custom Programming Language** | *IITGN* *Jan – Apr 2025*

- Engineered an end-to-end language toolchain comprising a recursive descent parser, lexical analyzer, and AST interpreter in Python for a dynamically typed, lexically scoped programming language.
- Built a back-end compiler and custom assembler targeting the **MIPS32 Instruction Set Architecture (ISA)**, translating high-level control flow and arithmetic expressions into low-level machine code.
- Implemented a lightweight Virtual Machine (VM) in C to execute compiled MIPS32 bytecode with custom memory management, register-file emulation, and an interactive shell interface.

## ACADEMIC AWARDS & ACHIEVEMENTS

- **HackRush'25 Hackathon:** Winner in the Computer Architecture track (held at IITGN).
- **Academic Excellence:** Dean's List for semesters 2, 4, 5, and 6.

## TECHNICAL SKILLS

**Hardware & Systems:** Verilog, SystemVerilog, C, Vivado, AXI4, DDR3 (MIG), UART, Ethernet (RMII/RGMII), Timing Analysis, Clock Domain Crossing (CDC), BRAM.

**Scientific Computing & ML:** Python, MATLAB, PyTorch, SciPy, NumPy, Pandas, OpenCV, SQL.

**Software Tools & Publishing:** LaTeX, Pandoc Markdown, Jupyter, Git, Bash, PowerShell, SSH.